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Research Notes

AN EARLY PATENT-LAW PROPOSAL?

L. SPRAGUE DE CAMP*

HISTORIANS OF TECHNOLOGY are familiar with the rise of patents and patent laws in Renaissance Italy. They have read how in 1421 Filippo Brunelleschi obtained from the Republic of Florence the first known patent, on a canal boat with cranes for handling heavy cargo; how in 1474 the Republic of Venice adopted the first patent law and in 1594 issued to Galileo a patent on a pump; how Galileo applied unsuccessfully for a number of other patents; and so on.¹

There is reason to suspect that the idea of patents for inventions was not new in Brunelleschi's time. In the eighth chapter of the second book of his *Politics*, Aristotle tells about

Hippodamos, the son of Euryphon, a native of Miletos, the same who invented the art of planning cities, and who also laid out the Peiraieus . . .²

and wrote the first known utopian tract on the constitution of the ideal city-state. Among Hippodamos' proposals,

He also enacted that those who discovered (*heuriskontôn*) anything for the good of the state should be honored . . .

or, in another translation,

He proposed a law that those who discovered something of advantage to the state should receive honor . . .

or,

And he proposed a law concerning those who discovered anything profitable for the city, that they should gain compensation . . .³

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¹ See, e. g., R. J. Forbes, *Man the Maker* (N. Y., 1958), p. 108; Friedrich Klemm, *A History of Western Technology* (N. Y., 1959), p. 171; Lewis Mumford, *Technics and Civilization* (N. Y., 1934), p. 132; W. B. Parsons: *Engineers and Engineering in the Renaissance* (Baltimore, 1939), p. 511.

² Aristotle, *Politics*, II, 8 (1267b).

³ *Ibid.*, (1268a). These translations are, first, by B. Jowett (1885); second, H.

This sounds like a proposal for a patent law, or at least for an official system of rewards for inventions. The verb *heuriskô*, although it also means “find out” and “obtain,” was the common Greek word for “invent, devise” in the technological sense. There were other verbs for “invent”—for example, *exeuriskô* (a derived form), *ktizô* (meaning primarily “originate, initiate”), *mêchanaomai*, *technaomai*, *tektainomai*, and *mêdomai*. But *heuriskô* remained the usual, commonplace term.

It would seem that Aristotle, reading Hippodamos’ treatise, took the latter’s proposal in a strictly political, non-technological sense. Further on, Aristotle objects to it on the ground that “it may encourage informers, or perhaps even lead to political commotions.”⁴ Then he enters into a discussion of whether laws ought ever to be changed, and if so when.

Because Aristotle is our only source for Hippodamos’ proposal, we cannot with any certainty go back of Aristotle to infer what Hippodamos really meant. We do know, however, that Hippodamos was a man of strong technological interests. He applied the grid pattern of streets, long known in Mesopotamia, not only to Peiraieus but also to Thourioi in Italy. He laid out the streets of Rhodes, built in a natural bowl, like the aisles and seats of a theater.

On the other hand, while Aristotle had as wide a range of interests as any mortal man can have, his interest in mechanics was, let us say, marginal. How strong we should think it was depends on the extent to which we believe that Aristotle was responsible for the *Mêchanika* included in his “minor works.” The authorship of this little treatise—the first known engineering textbook—is uncertain. One opinion is that, while the work may well contain an Aristotelian substratum, most of it is the creation of a later peripatetic, probably Straton of Lampsakos.⁵

The fact that the *Mêchanika* mentions the bronze purifying wheels mounted in Egyptian temples implies Straton’s authorship, at least in part. For such wheels, whose existence in Egypt is confirmed by Philon of Byzantium and Heron of Alexandria, are not mentioned in connection with Greek temples; Aristotle never went to Egypt, but Straton lived for years in Alexandria as tutor to the crown prince, later Ptolemaios II Philadelphos.⁶

Whatever Hippodamos meant, it seems likely that Aristotle, when he wrote his commentary, was thinking so hard about politics that the technological aspects of the proposal never occurred to him. And,

Rackham (1932); and the third, for this article, by Prof. William C. McDermott of the University of Pennsylvania. (Spelling normalized.)

⁴ *Ibid.*, (1268b).

⁵ See George Sarton, *Introduction to the History of Science* (Cambridge, Mass., 1959), Vol. I, p. 132; A. G. Drachmann, *The Mechanical Technology of Greek and Roman Antiquity* (Copenhagen, 1963), p. 10.

⁶ Aristotle, *Mechanics* (848a); Philon, *Pneumatics* (Carra de Vaux transl., Paris, 1903) lxiii; Heron, *Pneumatics* (Woodcroft transl., London, 1851), xxxi, lxviii.

although we cannot prove that Hippodamos had something like a patent law in mind, the possibility is, I think, strong enough to be worth mentioning in discussions of the origin of the theory of rewarding inventors. For, if any single change in institutions could have started the Industrial Revolution in classical times instead of 1,500 to 2,000 years later, a sound patent law was probably it.

*RESEARCH INTO PEASANT WORK AND ECONOMY
(AGRARIAN ETHNOGRAPHY) IN THE
GERMAN DEMOCRATIC REPUBLIC*

WOLFGANG JACOBET*

Very little attention was paid to peasant work and economy in the academic research programs in German ethnology before 1945. Instead, problems connected with such expressions as the "folk soul" or "folk spirituality" were in the foreground, and in that connection interest in the so-called material aspects of folk culture was confined to the colorful world of folk costume, the manifold examples of folk art, and the utensils in common use. Parallel with this, the entire field of folklore, investigations into folk beliefs and traditional festivals during the year or in a lifetime, occupied a prominent place. In this way a romantic and, in many respects, idyllic picture of the German peasants was bound to arise, for it lacked the essential basis of investigation of the daily work of the peasants in the fields, in the house, and in the farmyard. Only the peasant-house and peasant field and settlement customs were considered worth detailed consideration.¹

This neglect of research into the material bases of peasant life is all the less easy to understand because a beginning had already been made comparatively early outside Germany in investigating the imple-

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¹ The author is preparing a detailed study of these scientific historical questions of "material" ethnology in Germany. Cf. Wolfgang Jacobet, "Zur Erforschung der bäuerlichen Arbeit und Wirtschaft in der deutschen Volkskunde," *Deutsches Jahrbuch für Volkskunde*, Vol. VIII (1962), pp. 303 ff.; Günter Wiegelmann, "Zur Erfassung bäuerlicher Sachgüter und Arbeitsverfahren. Möglichkeiten und Probleme eines Fragebogens," *Protokollmanuskript der "3. Arbeitstagung über Fragen des Atlas der deutschen Volkskunde im Institut für geschichtliche Landeskunde der Rheinlande an der Universität Bonn vom 27. April bis 29. April 1961,"* pp. 19 ff.